

Variable Flow Documentation

Actual notebook variable names, shapes, dependencies, and next-use mapping.

Learning Day 7 - Decision Trees for Customer Segmentation

Documentation generated from the finalized notebook and project results.

Variable Flow Overview

This document tracks the actual notebook variables from raw Kaggle files through preprocessing, modeling, tuning, oblique comparison, and SHAP explanation. The goal is to make the notebook easier to debug, explain, and rerun.

Reading rule: each row shows what variable was created, what it was created from, what it contains, and where it is used next.

Variable Flow Map

Phase 1: Data Import and Preparation

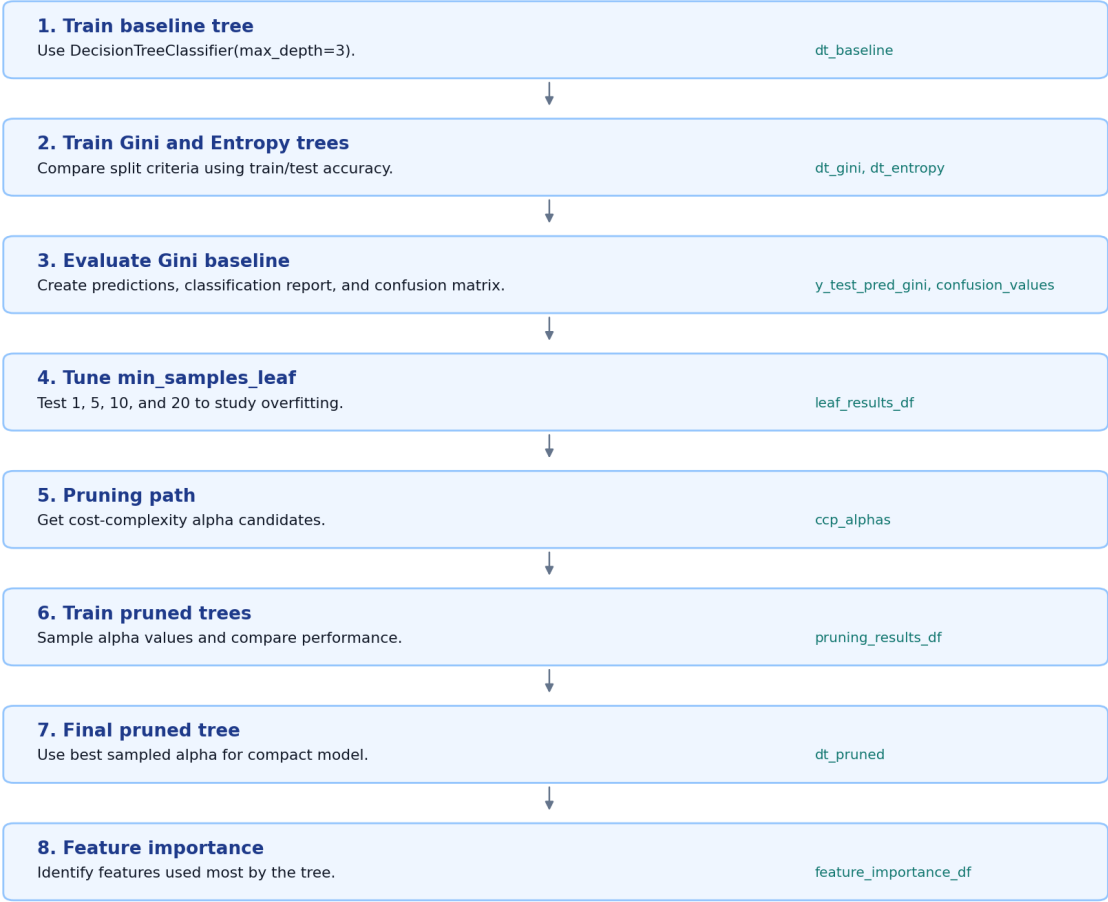
From raw Kaggle files to model-ready numeric feature matrices.



Data preparation variable flow.

Phase 2: Decision Tree Modeling and Tuning

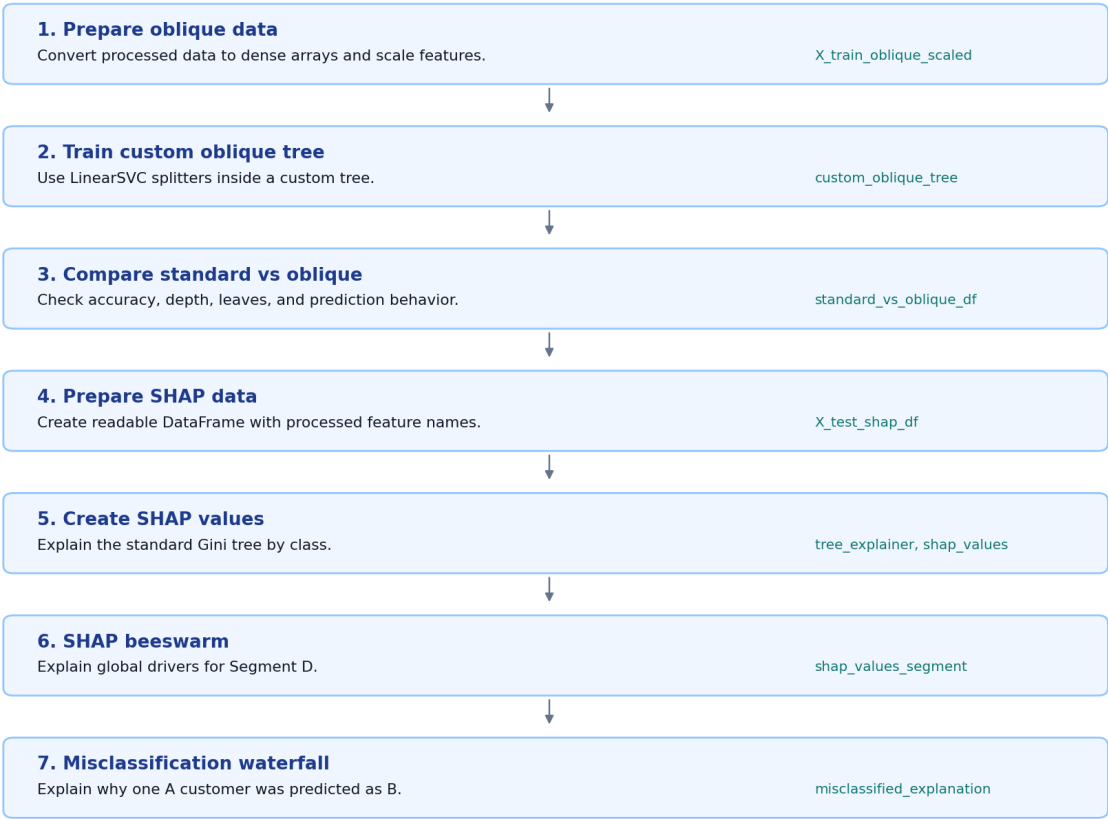
From baseline model to controlled, interpretable tree tuning.



Modeling and tuning variable flow.

Phase 3: Expert Comparison and Explainability

From expert oblique comparison to global and local explanation.



Expert comparison and explainability variable flow.

1. Data Import and Raw Files

Variable	Created from	Contains / shape	Used next
DATASET_SLUG	Manual assignment	kaushiksuresh147/customer-segmentation	kaggle download command
DATA_DIR	Path("data/customer_segmentation")	Project-relative data folder	train_path, test_path
train_path	DATA_DIR / "Train.csv"	Path to training CSV	pd.read_csv
test_path	DATA_DIR / "Test.csv"	Path to test CSV	pd.read_csv
train_df	pd.read_csv(train_path)	(8068, 11) raw training DataFrame	Data audit, X/y creation
test_df	pd.read_csv(test_path)	(2627, 11) raw test DataFrame	Initial inspection only
target_column	Manual assignment	Segmentation	X/y split, feature selection

2. Data Audit Variables

Variable	Created from	Contains / shape	Used next
num_rows, num_columns	train_df.shape	8068 rows, 11 columns	Dataset overview output
duplicate_count	train_df.duplicated().sum()	0 duplicates	Data quality note
missing_summary	train_df.isna().sum()	Missing counts for all columns	missing_summary_with_values
missing_summary_with_values	missing_summary filter	Only columns with missing values	Missing value interpretation
target_distribution	train_df["Segmentation"].value_counts()	Segment counts and shares	Target balance chart
id_column	Manual assignment	ID	Dropped from X
feature_df	train_df.drop([ID, Segmentation])	Raw feature-only DataFrame	numeric/categorical feature split
numeric_features	feature_df.select_dtypes	Age, Work_Experience, Family_Size	numeric_pipeline
categorical_features	feature_df.select_dtypes	Gender, Ever_Married, Graduated, Profession, Spending_Score, Var_1	categorical_pipeline

3. Feature and Target Variables

Variable	Created from	Contains / shape	Used next
X	train_df.drop([ID, Segmentation])	(8068, 9) raw input features	train_test_split
y	train_df["Segmentation"]	(8068,) target labels A/B/C/D	train_test_split
X_train	train_test_split(X, y)	(6454, 9) training features	preprocessor.fit_transform
X_test	train_test_split(X, y)	(1614, 9) test features	preprocessor.transform
y_train	train_test_split(X, y)	(6454,) training labels	model.fit
y_test	train_test_split(X, y)	(1614,) test labels	accuracy, reports, SHAP misclassification

4. Preprocessing Variables

Variable	Created from	Contains / shape	Used next
numeric_pipeline	Pipeline(SimpleImputer median)	Numeric missing-value imputation	preprocessor
categorical_pipeline	Pipeline(SimpleImputer mode + OneHotEncoder)	Categorical imputation and encoding	preprocessor
preprocessor	ColumnTransformer	Combined numeric and categorical preprocessing	fit_transform / transform
X_train_processed	preprocessor.fit_transform(X_train)	(6454, 28) processed training matrix	All model training
X_test_processed	preprocessor.transform(X_test)	(1614, 28) processed test matrix	All model evaluation
processed_numeric_features	numeric_features	Original numeric names	processed_feature_names
encoded_categorical_features	OneHotEncoder.get_feature_names_out	One-hot encoded categorical names	processed_feature_names
processed_feature_names	numeric + encoded names	28 feature names matching processed matrices	plots, SHAP, feature importance

5. Baseline and Evaluation Variables

Variable	Created from	Contains / shape	Used next
dt_baseline	DecisionTreeClassifier(max_depth=3)	Baseline decision tree	accuracy, visualization
dt_gini	DecisionTreeClassifier(criterion="gini")	Selected Gini baseline model	evaluation, SHAP
dt_entropy	DecisionTreeClassifier(criterion="entropy")	Entropy comparison model	criterion comparison
y_train_pred_baseline	dt_baseline.predict(X_train_processed)	Training predictions	baseline accuracy
y_test_pred_baseline	dt_baseline.predict(X_test_processed)	Test predictions	baseline accuracy
train_accuracy_baseline	accuracy_score(y_train, y_train_pred_baseline)	0.4930	comparison tables
test_accuracy_baseline	accuracy_score(y_test, y_test_pred_baseline)	0.5105	comparison tables
y_train_pred_gini	dt_gini.predict(X_train_processed)	Gini train predictions	classification report
y_test_pred_gini	dt_gini.predict(X_test_processed)	Gini test predictions	classification report, confusion, SHAP error analysis
confusion_values	confusion_matrix(y_test, y_test_pred_gini)	4x4 confusion matrix	heatmap and TP/FP/FN/TN
confusion_matrix_df	pd.DataFrame(confusion_values)	Readable confusion matrix	display
tp_fp_fn_tn_df	loop over confusion_values	One-vs-rest TP/FP/FN/TN table	interpretation

6. Tuning and Pruning Variables

Variable	Created from	Contains / shape	Used next
leaf_values	Manual list	[1, 5, 10, 20]	min_samples_leaf experiment
leaf_results	Loop output list	Metric dictionaries for each leaf size	leaf_results_df
leaf_results_df	pd.DataFrame(leaf_results)	leaf tuning results	accuracy chart
pruning_model	DecisionTreeClassifier(gini)	Unpruned tree used for pruning path	cost_complexity_pruning_path
pruning_path	pruning_model.cost_complexity_pruning_path	Alpha path object	ccp_alphas
ccp_alphas	pruning_path.ccp_alphas	1,476 alpha candidates	sampled_alphas
sampled_alphas	np.linspace(min, max, 25)	25 sampled alpha values	pruned tree loop
pruning_results	Loop output list	Metric dictionaries for alphas	pruning_results_df
pruning_results_df	pd.DataFrame(pruning_results)	Pruning comparison table	chart and best alpha
best_pruning_row	idxmax test accuracy	Best sampled pruning row	best_ccp_alpha
best_ccp_alpha	best_pruning_row["ccp_alpha"]	0.002601	dt_pruned

Variable	Created from	Contains / shape	Used next
dt_pruned	DecisionTreeClassifier(ccp_alpha=best_ccp_alpha)	Final pruned tree	baseline_vs_pruned_df
baseline_vs_pruned_df	Manual DataFrame	Baseline vs pruned metrics	model comparison

7. Feature Importance Variables

Variable	Created from	Contains / shape	Used next
feature_importance_values	dt_gini.feature_importances_	Importance value per processed feature	feature_importance_df
feature_importance_df	DataFrame feature + importance	Non-zero feature importance table	feature importance chart
chart_data	feature_importance_df sorted	Chart-ready importance data	horizontal bar chart

8. Custom Oblique Tree Variables

Variable	Created from	Contains / shape	Used next
ObliqueTreeNode	dataclass	One node in custom oblique tree	CustomObliqueDecisionTree
CustomObliqueDecisionTree	Custom class	Classifier with fit/predict/depth/leaves methods	custom_oblique_tree
X_train_oblique	dense conversion of X_train_processed	(6454, 28) dense float input	StandardScaler
X_test_oblique	dense conversion of X_test_processed	(1614, 28) dense float input	StandardScaler
oblique_scaler	StandardScaler()	Scaler fitted on training features	X_train_oblique_scaled
X_train_oblique_scaled	oblique_scaler.fit_transform(X_train_oblique)	(6454, 28) scaled training features	custom_oblique_tree.fit
X_test_oblique_scaled	oblique_scaler.transform(X_test_oblique)	(1614, 28) scaled test features	custom_oblique_tree.predict
label_encoder_oblique	LabelEncoder()	Maps A/B/C/D to 0/1/2/3	y_train_oblique, inverse transform
y_train_oblique	label_encoder_oblique.fit_transform(y_train)	(6454,) encoded train labels	custom_oblique_tree.fit
y_test_oblique	label_encoder_oblique.transform(y_test)	(1614,) encoded test labels	accuracy_score
custom_oblique_tree	CustomObliqueDecisionTree(...)	Trained oblique-style classifier	comparison and predictions
y_test_pred_custom_oblique	custom_oblique_tree.predict(X_test_oblique_scaled)	Encoded oblique test predictions	accuracy and distribution
standard_vs_oblique_df	Manual DataFrame	Standard vs oblique model comparison	expert conclusion
custom_oblique_prediction_distribution	value_counts of predictions	Predicted segment distribution	bias check

9. SHAP Variables

Variable	Created from	Contains / shape	Used next
X_test_shap	dense conversion of X_test_processed	(1614, 28) SHAP input array	X_test_shap_df
X_test_shap_df	pd.DataFrame(X_test_shap, columns=processed_feature_names)	(1614, 28) SHAP DataFrame	sample, waterfall
X_test_shap_sample	X_test_shap_df.sample(n=300)	(300, 28) sampled SHAP data	tree_explainer
tree_explainer	shap.TreeExplainer(dt_gini)	SHAP explainer for Gini tree	shap_values
shap_values	tree_explainer(X_test_shap_sample)	values shape (300, 28, 4)	beeswarm
segment_to_explain	Manual assignment	D	segment_index
segment_index	dt_gini.classes_.index("D")	Class index for Segment D	shap_values_segment

Variable	Created from	Contains / shape	Used next
shap_values_segment	shap.Explanation single class	SHAP values for Segment D only	beeswarm plot
mean_abs_shap_values	np.abs(shap_values).mean	Mean absolute SHAP per feature	shap_importance_df
shap_importance_df	DataFrame feature + mean_abs_shap	BI feature impact table	top_shap_features
prediction_check_df	y_test vs y_test_pred_gini	Actual/predicted comparison table	misclassified_df
misclassified_df	filter incorrect rows	Wrong predictions only	misclassified_position
misclassified_position	misclassified_df.index[0]	0	waterfall
actual_segment	prediction_check_df.loc[position]	A	waterfall title
predicted_segment	prediction_check_df.loc[position]	B	waterfall class
misclassified_customer_df	X_test_shap_df.iloc[[position]]	One customer row	misclassified_shap_values
misclassified_shap_values	tree_explainer(misclassified_customer_df)	SHAP for one customer	official and BI waterfall
waterfall_df	feature/value/effect DataFrame	Readable SHAP effects table	top_waterfall_df
top_waterfall_df	top 8 effects	BI-ready waterfall data	labelled waterfall chart

End-to-End Variable Dependency Summary

Phase	Key source variables	Key output variables
Raw data	DATA_DIR, train_path, test_path	train_df, test_df
Feature setup	train_df, id_column, target_column	X, y, numeric_features, categorical_features
Split	X, y	X_train, X_test, y_train, y_test
Preprocessing	preprocessor, X_train, X_test	X_train_processed, X_test_processed, processed_feature_names
Baseline tree	X_train_processed, y_train	dt_baseline, dt_gini, y_test_pred_gini
Tuning	dt_gini, leaf_values, ccp_alphas	leaf_results_df, pruning_results_df, dt_pruned
Oblique comparison	X_train_processed, y_train, X_test_processed, y_test	custom_oblique_tree, standard_vs_oblique_df
SHAP	dt_gini, X_test_shap_df, y_test_pred_gini	shap_values, shap_values_segment, waterfall_df